

NANOFERTILIZER

Formulation, Suspension Handling and Plant Evaluation

Technical research note based on the formulation discussion

Current formulation Nano-MgO, nano-CaCO₃, nano-ZnO, nano-MnO₂ and nano-CuO maintained as a 20 wt% aqueous suspension. HAP is excluded. Urea is stored separately and combined only after dilution immediately before application.

Critical limitation The proposed ratios and concentrations are experimental starting points, not validated field recommendations. Crop, soil, nanoparticle size, surface chemistry and dissolution must be tested before agricultural use.

1. Executive summary

The proposed material can be evaluated as a calcium-magnesium and micronutrient nano-amendment. It is not a complete fertilizer by itself because it supplies no meaningful nitrogen, phosphorus or potassium. Urea can provide nitrogen, while phosphorus and potassium must be supplied separately when soil testing shows a requirement.

Recommended product architecture Use a two-part system: Part A is the 20 wt% nanoparticle suspension; Part B is a crop-specific urea solution. Combine them after dilution and use immediately. Do not store ammonium nitrate with the CuO-containing nano-suspension.

2. Final nano-premix composition

Because every powder is nanoscale, the biologically active metal oxides are kept low. The following dry-solids ratio is a conservative starting point for controlled screening.

Nanopowder	Dry-solids fraction	Role
Nano-MgO	50.0 wt%	Magnesium source; chlorophyll and enzyme function
Nano-CaCO ₃	42.8 wt%	Calcium source and liming contribution
Nano-ZnO	5.0 wt%	Zinc micronutrient source
Nano-MnO ₂	2.0 wt%	Manganese micronutrient source
Nano-CuO	0.2 wt%	Copper micronutrient source; intentionally minimized

Equivalent mass ratio: MgO : CaCO₃ : ZnO : MnO₂ : CuO = 250 : 214 : 25 : 10 : 1.

3. Nutritional function and limitations

Material	Potential benefit	Principal concern
MgO	Mg for chlorophyll, photosynthesis and enzymes	May raise rhizosphere pH; dose-dependent phytotoxicity
CaCO ₃	Ca for cell walls; neutralizes acidic soil	May suppress Zn and Mn availability in neutral/alkaline soil
ZnO	Zn for enzymes, protein synthesis and biofortification	Excess Zn can cause oxidative stress
MnO ₂	Mn for photosystem II and antioxidant enzymes	Excess Mn can inhibit growth
CuO	Cu for redox enzymes, lignification and defense	Narrow beneficial-to-toxic range; microbial effects

Soil-pH constraint The high MgO and CaCO₃ fraction makes this concept most relevant to acidic soils. For neutral or alkaline soils, redesign the carrier balance instead of assuming the same ratio will remain effective.

4. The 20 wt% suspension

A 20 wt% suspension contains 200 mg of nanoparticle solids per gram of suspension. It is a concentrated stock, not a direct-application liquid. Gravimetric dosing is preferred because suspension density and sedimentation make volume-only dosing less reliable.

Stock mass (g) = target nano concentration (mg/L) x final volume (L) / 200.

Target total nanoparticles	Stock per 1 L	Approx. dilution
5 mg/L	0.025 g	1:40,000
10 mg/L	0.050 g	1:20,000
25 mg/L	0.125 g	1:8,000
50 mg/L	0.250 g	1:4,000

5. Experimental dose matrix

The table below converts the total-premix treatments into the concentration of each nanopowder. Begin with the low range and advance only when germination, root-growth and oxidative-stress endpoints remain acceptable.

Total premix	MgO	CaCO ₃	ZnO	MnO ₂	CuO
5 mg/L	2.50	2.14	0.25	0.10	0.01
10 mg/L	5.00	4.28	0.50	0.20	0.02
25 mg/L	12.50	10.70	1.25	0.50	0.05
50 mg/L	25.00	21.40	2.50	1.00	0.10

All component values in the table are mg/L. These concentrations are screening treatments, not field-use recommendations.

6. Combining the suspension with urea

Preferred approach Keep the nano-suspension and urea separate during storage. Dissolve urea first, add the freshly redispersed nano-stock after dilution, mix continuously and apply immediately as a soil drench.

1. Add approximately 80% of the required deionized water to a clean mixing vessel.
2. Dissolve the crop-specific quantity of urea completely under stirring.
3. Redisperse the 20 wt% nano-stock by shaking or controlled low-energy sonication.
4. Weigh the required nano-stock; do not rely on volume unless stock density has been measured.
5. Add the nano-stock slowly to the diluted urea solution while stirring.
6. Bring the mixture to final volume, record pH and conductivity, and continue agitation.
7. Apply immediately as a soil drench. Do not retain the mixed solution for later use.

7. Urea calculation

Commercial urea contains approximately 46% nitrogen. Calculate the urea amount from the nitrogen requirement rather than from the nanoparticle dose.

Required urea mass = required nitrogen mass / 0.46.

Example: supplying 1.00 g nitrogen requires approximately 2.17 g urea. The correct pot or field quantity must be derived from the crop recommendation, soil mass, target area and application schedule.

Example nano addition For 1 L containing 25 mg/L total nanoparticles, add 0.125 g of the 20 wt% nano-stock to the already diluted urea solution. This supplies 12.5 mg/L MgO, 10.7 mg/L CaCO₃, 1.25 mg/L ZnO, 0.50 mg/L MnO₂ and 0.05 mg/L CuO.

8. Why a stored urea-nanoparticle suspension is not preferred

- Dissolved urea raises ionic strength and may destabilize nanoparticle surface charge.
- MgO and CaCO₃ can change pH, dissolution and urea-related ammonia losses.
- ZnO, MnO₂ and CuO may aggregate or adsorb to the larger mineral fraction.
- Sedimentation can produce a nonuniform nutrient dose.
- Adding urea changes the nominal 20 wt% nanoparticle concentration.

If a single stored product is required, it should be treated as a separate formulation-development project involving dispersant selection, rheology, accelerated storage stability, redispersibility, dissolution and crop-safety testing.

9. Ammonium nitrate warning

Do not store together Ammonium nitrate is an oxidizer, and copper contamination is a recognized decomposition concern. Do not prepare or store a homemade ammonium-nitrate/urea/nano-CuO concentrate. If nitrate nitrogen is needed, use a commercial fertilizer-grade product as a separate treatment under institutional safety procedures.

10. Greenhouse experimental design

Use a randomized pot study with at least four biological replicates. A practical treatment structure is:

- Untreated control.
- Urea-only control at the recommended nitrogen dose.
- Urea plus 5, 10, 25 and 50 mg/L total nano-premix.
- Bulk, non-nano mineral mixture with the same elemental composition.
- Individual nanoparticle treatments to identify the component driving benefit or toxicity.
- A conventional soluble micronutrient control matched for elemental Cu, Zn and Mn.

Initially use soil application rather than foliar spraying. The suspension contains poorly soluble mineral particles that may settle, block spray equipment or create localized leaf exposure.

11. Measurements required

11.1 Suspension and material characterization

- XRD for crystalline phases and phase stability.
- TEM for primary particle size and morphology.
- SEM-EDS for elemental distribution.
- DLS for hydrodynamic size in stock and final treatment water.
- Zeta potential, pH, conductivity, viscosity and sedimentation.
- ICP-OES or ICP-MS for total and dissolved metals.
- Dissolution and release in water and representative soil extracts.

11.2 Plant and soil endpoints

- Germination percentage, root length and shoot length.
- Fresh and dry biomass, chlorophyll/SPAD and photosynthetic efficiency.
- Plant-tissue Cu, Zn, Mn, Mg and Ca.
- Oxidative-stress markers and SOD, CAT and POD activities.
- Soil pH, available nutrients, respiration and enzyme activity.
- Yield, nutrient-use efficiency and edible-tissue accumulation where relevant.

12. Interpretation rules

A nanoparticle treatment should not be considered successful solely because shoot length increases. A defensible conclusion requires improved biomass or yield without excessive tissue-metal accumulation, oxidative stress, root injury, adverse soil-microbial effects or unfavorable soil-pH change.

Key scientific distinction Physical mixing creates a nanoparticle blend or suspension. It should not be called a chemically integrated nanocomposite unless microscopy, spectroscopy, diffraction or surface analysis demonstrates interfacial integration or a new composite structure.

13. Recommended next decision

Before selecting a final formulation or field rate, determine the target crop, soil pH, soil texture, organic matter, existing Mg/Ca/Zn/Mn/Cu status, primary particle sizes, hydrodynamic sizes, and dissolved-metal release. Those data should drive the next ratio adjustment.

14. Research evidence and links

- [Nanofertilizers in Modern Agriculture: A Technological Revolution in Plant Nutrition and Resource Efficiency](#)
- [A Comprehensive Review on Nano-Fertilizers: Preparation, Properties and Applications](#)
- [Nanofertilizers: A Smart and Sustainable Attribute to Modern Agriculture](#)
- [Development and Testing of Zinc Sulfate and Zinc Oxide Nanoparticle-Coated Urea](#)
- [New Slow-Release Urea Fertilizer Fortified with Zinc for Improved Wheat Performance](#)
- [Facile Coating of Micronutrient Zinc for Slow-Release Urea](#)
- [Zinc Oxide Nanofertilizer Effects on Wheat Growth, Yield and Biofortification](#)
- [Foliar Spraying of MnO₂ Nanoparticles and Effects on Mung Bean](#)

- [Manganese Oxide Nanoparticles as a Seed-Priming Agent](#)
- [Comparative Effects of Manganese Nanoparticles and Conventional Forms](#)
- [Inhibitory Impact of MgO Nanoparticles on Oxidative Stress and Crop Response](#)
- [Impact of Metal and Metal Oxide Nanoparticles on Plants](#)
- [Biosynthesized Metal Oxide Nanoparticles for Sustainable Agriculture](#)
- [Safe Storage, Handling and Management of Ammonium Nitrate](#)
- [FM Global: Ammonium Nitrate and Mixed Fertilizers](#)